Sampling, Nyquist and the ADC

IoT · University · Lecture 1



In this lecture

- State and apply the sampling theorem.
- Predict aliases.
- Size ADC resolution.

Remember

- fs > 2B.
- Filter before sampling.
- step = V_ref / 2^n.

Predict the alias

1. Warm-up (5 min): Why backwards?
2. Teach (15 min): The theorem.
3. Predict (15 min): Five rates.
4. Measure (20 min): On hardware.
5. Wrap-up (5 min): One-minute paper.

Let’s discuss

1. Which prediction failed, and why?
2. Why can’t software undo aliasing?
3. How many bits does your sensor need?

Exit ticket

- Give the alias of a 90 Hz tone sampled at 100 Hz.